6

C.D.U./U.D.C. 621.316.17.016.34-756.001.25 614.825

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

NORME DE LA CEI

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC STANDARD

Publication 364-4-473

Première édition — First edition

1977

Installations électriques des bâtiments

Quatrième partie: Protection pour assurer la sécurité

Chapitre 47: Application des mesures de protection pour assurer la sécurité

Section 473 — Mesures de protection contre les surintensités

Electrical installations of buildings

Part 4: Protection for safety

Chapter 47: Application of protective measures for safety

Section 473 — Measures of protection against overcurrent

Bureau Central de la Commission Electrotechnique Internationale

1, rue de Varembé

Genève, Suisse

Prix
Price Fr. s. 16.—

INTERNATIONAL ELECTROTECHNICAL COMMISSION

ELECTRICAL INSTALLATIONS OF BUILDINGS

Part 4: Protection for safety

Chapter 47: Application of protective measures for safety

Section 473 — Measures of protection against overcurrent

FOREWORD

1) The formal decisions or agreements of the I E C on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
3) In order to promote international unification, the I E C expresses the wish that all National Committees should adopt the text of the I E C recommendation for their national rules in so far as national conditions will permit. Any divergence between the I E C recommendation and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This standard has been prepared by I E C Technical Committee No. 64, Electrical Installations of Buildings.

The requirements of this standard form a complement to the rules contained in Chapter 43, Protection against overcurrent (I E C Publication 364-4-43).

A draft of this standard was discussed at the meeting held in Toronto in 1976. As a result of this meeting, a draft, Document 64(Central Office)51, was submitted to the National Committees for approval under the Six Month's Rule in September 1976.

The following countries voted explicitly in favour of publication:

Canada
Denmark
Egypt
France
Germany
Israel
Italy
Japan
Netherlands
Poland
Portugal
Romania
South Africa (Republic of)
Sweden
Switzerland
Turkey
United States of America

The British Committee voted against this standard because it cannot accept the conditions of Sub-clause 473.2.2 which it considers to be too lenient and possibly dangerous.

ELECTRICAL INSTALLATIONS OF BUILDINGS

Part 4: Protection for safety

Chapter 47: Application of protective Measures for safety

Section 473 — Measures of protection against overcurrent

---

473. MEASURES OF PROTECTION AGAINST OVERCURRENT

Note. — The requirements of this section do not take account of external influences. For the application of protective measures in relation to conditions of external influences, see Chapter 48 (under consideration).

473. Measures of protection against overcurrent

473.1.1 *Position of devices for overload protection*

473.1.1.1 A device ensuring protection against overload shall be placed at the point where a change, such as a change in cross-sectional area, nature, method of installation or in constitution, causes a reduction in the value of current-carrying capacity of the conductors, with the exceptions mentioned in Sub-clauses 473.1.1.2 and 473.1.2.

473.1.1.2 The device protecting the wiring against overload may be placed along the run of that wiring if the part of the wiring between the point where a change occurs (in cross-sectional area, nature, method of installation or constitution) and the position of the protective device has neither branch circuits nor socket-outlets and fulfils one of the following two conditions:

a) it is protected against short-circuit current in accordance with the requirements stated in Section 434;

b) its length does not exceed 3 m, it is carried out in such a manner as to reduce the risk of short circuit to a minimum, and it is not placed near combustible material (see Sub-clause 473.2.2.1).

473.1.2 *Omission of devices for protection against overload*

The various cases stated in this sub-clause shall not be applied in installations situated in locations presenting a fire risk or risk of explosion and where special rules for certain locations specify different conditions.

Devices for protection against overload need not be provided for:

a) wiring situated on the load side of a change in cross-sectional area, nature, method of installation or in constitution, and effectively protected against overload by a protective device placed on the supply side;

b) wiring which is not likely to carry overload current, on condition that this wiring shall be protected against short circuit in accordance with the requirements of Section 434 and that it has neither branch circuits nor socket-outlets;

c) installations for telecommunications, control, signalling and the like.

Note. — Conditions for overload protection for the installations mentioned in item *c)* are under consideration.

473.1.3 *Position or omission of devices for protection against overload in IT systems*

The provisions in Sub-clauses 473.1.1.2 and 473.1.2 for alternative position or omission of devices for protection against overload are not applicable to IT systems unless each circuit not protected against overload is protected by a residual current-operated protective device, or all the equipment supplied by such a circuit—including the wiring—is carried out according to the protective measure described in Clause 413.2.

473.1.4 *Cases where omission of devices for overload protection is recommended for safety reasons*

The omission of devices for protection against overload is recommended for circuits supplying current-using equipment where unexpected opening of the circuit could cause danger.

Examples of such cases are:

— exciter circuits of rotating machines;

— supply circuits of lifting magnets;

— secondary circuits of current transformers.

Note. — In such cases consideration should be given to the provision of an overload alarm.

473.2 **Protection against short circuit**

473.2.1 *Position of devices for short-circuit protection*

A device ensuring protection against short circuit shall be placed at a point where a reduction in the cross-sectional area of the conductors or another change causes modification of the characteristics specified in Sub-clause 473.1.1.1, except where Sub-clauses 473.2.2 or 473.2.3 apply.

473.2.2 *Alternative position of devices for short-circuit protection*

It is permissible to provide devices for protection against short circuit at a place other than that specified in Sub-clause 473.2.1, under the conditions stated in Sub-clauses 473.2.2.1 or 473.2.2.2.

473.2.2.1 The part of the wiring between the point of reduction of cross-sectional area or other change and the position of the protective device fulfils simultaneously, the three following conditions:

a) its length does not exceed 3 m;

b) it is carried out in such a manner as to reduce the risk of a short circuit to a minimum;

Note. — This condition may be obtained for example by reinforcing the protection of the wiring against external influences.

c) it is not placed close to combustible material.

473.2.2.2 A protective device placed on the supply side of the reduced cross-sectional area or other change possesses an operating characteristic such that it protects the wiring situated on the load side against short circuit, in accordance with the rule of Sub-clause 434.3.2.

473.2.3 *Omission of devices for short circuit protection*

Devices for protection against short circuit need not be provided for:

— conductors connecting generators, transformers, rectifiers, accumulator batteries to the associated control panels, the protective devices being placed on these panels;

— circuits where disconnection could cause danger for the operation of the installations concerned, such as those quoted in Sub-clause 473.1.4;

— certain measuring circuits;

provided that the two following conditions are simultaneously fulfilled:

a) the wiring is carried out in such a way as to reduce the risk of a short circuit to a minimum (see item *b)* of Sub-clause 473.2.2.1);

b) the wiring shall not be placed close to combustible material.

473.2.4 *Short-circuit protection for conductors in parallel*

A single device may protect several conductors in parallel against short circuit provided that the operating characteristics of the device and the method of installation of the parallel conductors are suitably co-ordinated; for selection of the protective device see Chapter 53 (under consideration).

473.3 **Requirements according to the nature of the circuits**

473.3.1 *Protection of phase conductors*

473.3.1.1 Detection of overcurrent shall be provided for all phase conductors; it shall cause the disconnection of the conductor in which the overcurrent is detected, but not necessarily the disconnection of other live conductors, except where Sub-clause 473.3.1.2 applies.

473.3.1.2 In TT systems, for circuits supplied between phases and in which the neutral conductor is not distributed, overcurrent detection need not be provided for one of the phase conductors, provided that the following conditions are simultaneously fulfilled:

a) there exists, in the same circuit or on the supply side, differential protection intended to cause disconnection of all the phase conductors;

b) the neutral conductor is not distributed from an artificial neutral point of the circuits situated on the load side of the differential protective device mentioned in *a)*.

Note. — If disconnection of a single phase may cause danger, for example in the case of three-phase motors, appropriate precautions shall be taken.

473.3.2 *Protection of the neutral conductor*

473.3.2.1 *TT or TN systems*

a) Where the cross-sectional area of the neutral conductor is at least equal or equivalent to that of the phase conductors, it is not necessary to provide overcurrent detection for the neutral conductor or a disconnecting device for that conductor.

b) Where the cross-sectional area of the neutral conductor is less than that of the phase conductors, it is necessary to provide overcurrent detection for the neutral conductor, appropriate to the cross-sectional area of that conductor; this detection shall cause the disconnection of the phase conductors, but not necessarily of the neutral conductor.

However, overcurrent detection need not be provided for the neutral conductor if the two following conditions are simultaneously fulfilled:

— the neutral conductor is protected against short circuit by the protective device for the phase conductors of the circuit, and

— the maximum current likely to traverse the neutral conductor is, in normal service, clearly less than the value of the current-carrying capacity of that conductor.

Note. — This second condition is satisfied if the power carried is shared as evenly as possible between the different phases, for example if the sum of the powers absorbed by current-using equipment supplied from each phase and neutral (such as lighting and socket-outlets) is much less than the total power carried by the circuit concerned. The cross-sectional area of the neutral conductor should be not less than the appropriate value prescribed in Chapter 52 (under consideration).

473.3.2.2 *IT systems*

In IT systems it is strongly recommended that the neutral conductor should not be distributed.

However, where the neutral conductor is distributed, it is generally necessary to provide overcurrent detection for the neutral conductor of every circuit, which will cause the disconnection of all the live conductors of the corresponding circuit, including the neutral conductor. This measure is not necessary if:

— the particular neutral conductor is effectively protected against short circuit by a protective device placed on the supply side, for example at the origin of the installation, in accordance with the rules stated in Clause 434.3;

— or if the particular circuit is protected by a residual current operated protective device with a rated residual current not exceeding 0.15 times the current-carrying capacity of the corresponding neutral conductor. This device shall disconnect all the live conductors of the corresponding circuit, including the neutral conductor.

473.3.3 *Disconnection and reconnection of the neutral conductor*

Where disconnection of the neutral conductor is required, disconnection and reconnection shall be such that the neutral conductor shall not be disconnected before the phase conductors and shall be reconnected at the same time as or before the phase conductors.